

up this way for future expansion, as discussed in the note at the end of this section.) The balloon manager then uses the `remove()` method to eliminate that `Balloon` from its list, increments the number of popped balloons, and updates the score.

In each iteration of the main loop, the main code also calls the `update()` method of the balloon manager ❸, which passes this call on to all of the balloons, telling them to update themselves. Each balloon moves up the screen based on its own speed setting and returns its status: either that it is still moving (`BALLOON_MOVING`) or that it has moved beyond the top of the window (`BALLOON_MISSED`). If a balloon was missed, the balloon manager removes that balloon from its list and increments its count of missed balloons.

The balloon manager provides three getter methods that allow the main code to get the score ❹, the number of popped balloons ❺, and the number of missed balloons ❻.

Each time through the main loop, the main code calls the balloon manager's `draw()` method ❼. The balloon manager doesn't have anything to draw by itself, but loops through all the `Balloon` objects and calls the `draw()` method of each. (Notice the polymorphism here. The balloon manager has a `draw()` method, and each `Balloon` object has a `draw()` method.)

NOTE

As a challenge, try to expand this game to include a new type (subclass) of `Balloon`, a `MegaBalloon`. The difference is that a `MegaBalloon` will take three clicks to pop. Artwork is included in the download for this game.

Balloon Class and Objects

Finally, we have the balloon classes. To reinforce the concept of inheritance from Chapter 10, the `Balloon.py` module includes an abstract base class named `Balloon` and three subclasses: `BalloonSmall`, `BalloonMedium`, and `BalloonLarge`. The balloon manager instantiates `Balloon` objects from these subclasses. The subclasses each only include an `__init__()` method, which overrides and then calls the abstract method `__init__()` in the `Balloon` class. Each balloon image will start at some randomized location (below the